Perturbation theory

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Approximation method

Disterbance / Displacement / Difference

1st order / 2nd order

Importance:

- There are many example where Schrodinger eqn is not solved. So we can't get exact E.

Ex. Atoms with # of e > 1 (multielectron system)

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So, we apply approximation methods.

Non-Sci example: A soln of 3gm/Lt is given for Titration to a class. After experiment students returned with values of 2.8, 2.9, 3.1, 3.2, 3.3 g/Lt. So, difference in results are, −2, −1, +0.1 +0.2, +0.3 .....

Perturbation = Exact value − Real value

Example 2: wt = x g

wt. of headphone = 100g, So total wt. = (x+100) g

Perturbation = (x + 100) − x = 100 g

Derivation:

Schrodinger eqn, $\hat{H}\psi = E\psi$ —(1) for perturbed system

$\hat{H}^{(0)}\psi^{(0)} = E^{(0)}\psi^{(0)}$ —(2) un-perturbed system

Perturbation: $\Delta H = \hat{H} - \hat{H}^{(0)} \Rightarrow \hat{H} = \Delta H + \hat{H}^{(0)}$

$\Delta E = E - E^{(0)} \Rightarrow E = \Delta E + E^{(0)}$

$\Delta\psi = \psi - \psi^{(0)} \Rightarrow \psi = \Delta\psi + \psi^{(0)}$

Put this eqn in eqn (1)

$$(\Delta H + \hat{H}^{(0)})(\Delta\psi + \psi^{(0)}) = (\Delta E + E^{(0)})(\Delta\psi + \psi^{(0)})$$

$$\Rightarrow (\Delta H\Delta\psi + \Delta H\psi^{(0)} + \hat{H}^{(0)}\Delta\psi + \hat{H}^{(0)}\psi^{(0)}) = (\Delta E\Delta\psi + \Delta E\psi^{(0)} + E^{(0)}\Delta\psi + E^{(0)}\psi^{(0)})$$

Replace $\hat{H}^{(0)}\psi^{(0)} = E^{(0)}\psi^{(0)}$ in left side ⇒

$$\Delta H\Delta\psi + \Delta H\psi^{(0)} + \hat{H}^{(0)}\Delta\psi = \Delta E\Delta\psi + \Delta E\psi^{(0)} + E^{(0)}\Delta\psi$$

Neglecting $\Delta H\Delta\psi$, as it's value is very small ⇒

$$\Delta H\Delta\psi^{(0)} + \hat{H}\Delta\psi = \Delta E\psi^{(0)} + E^{(0)}\Delta\psi$$

$$\Rightarrow \Delta H\psi^{(0)} + \hat{H}^{(0)}\Delta\psi - E^{(0)}\Delta\psi = \Delta E\psi^{(0)}$$

$$\Rightarrow \Delta H\psi^{(0)} + (\hat{H}^{(0)} - E^{(0)})\Delta\psi = \Delta E\psi^{(0)}$$

Multiplying by $\psi^{(0)*}$ on both sides ⇒ and then integrating ⇒

$$\int_{-\infty}^{+\infty}\psi^{(0)*}\Delta H\psi^{(0)}d\tau + \underbrace{\int_{-\infty}^{+\infty}\psi^{(0)*}(\hat{H}^{(0)} - E^{(0)})\Delta\psi\, d\tau}_{=0\ \text{(zero)}} = \int_{-\infty}^{+\infty}\psi^{(0)*}\Delta E\psi^{(0)}d\tau$$

Because Hermitian operator

$$= \Delta E\underbrace{\int_{-\infty}^{+\infty}\psi^{(0)*}\psi^{(0)}d\tau}_{=1}$$

=1 as $\psi^{(0)}$ is normalized wave function

$$= \Delta E$$

Finally,

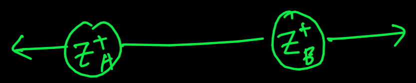
$$\boxed{\Delta E = \int_{-\infty}^{+\infty}\psi^{(0)*}\Delta H\psi^{(0)}d\tau}$$

We know, $\Delta E = E - E^{(0)}$, or, $E = \Delta E + E^{(0)}$

$$\boxed{\text{So, } E = \int_{-\infty}^{+\infty}\psi^{(0)*}\Delta H\psi^{(0)}d\tau + E^{(0)}}$$

Assignment — 1

LCAO – MO Theory



Internuclear Axis

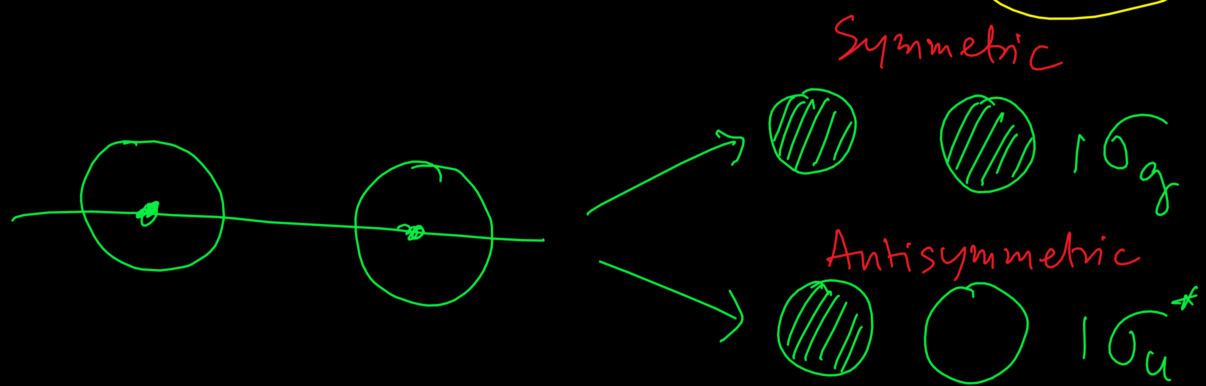
Homo nuclear Diatomics $Z_A = Z_B$

Period 1 : H_2, He_2 1s

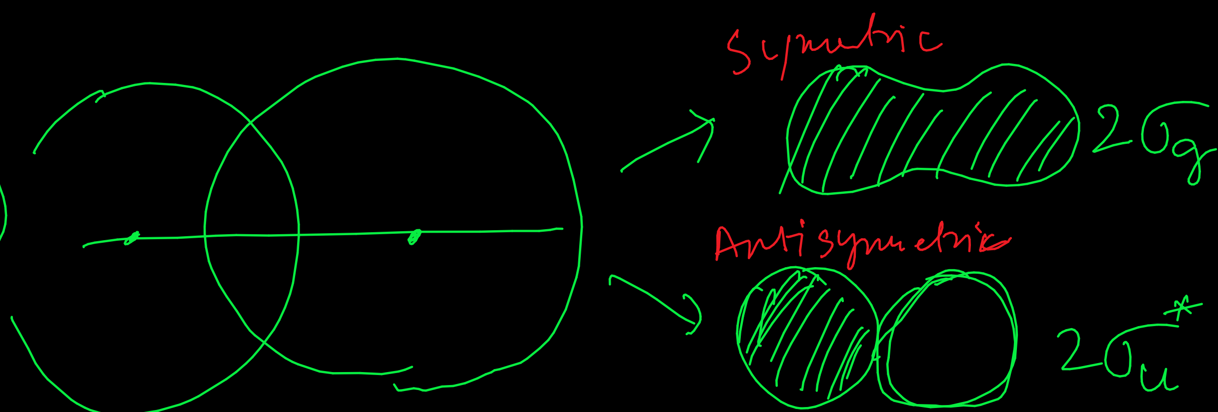
Period 2 : Li_2, Be_2, B_2, C_2, O_2, F_2, Ne_2

2s 2p

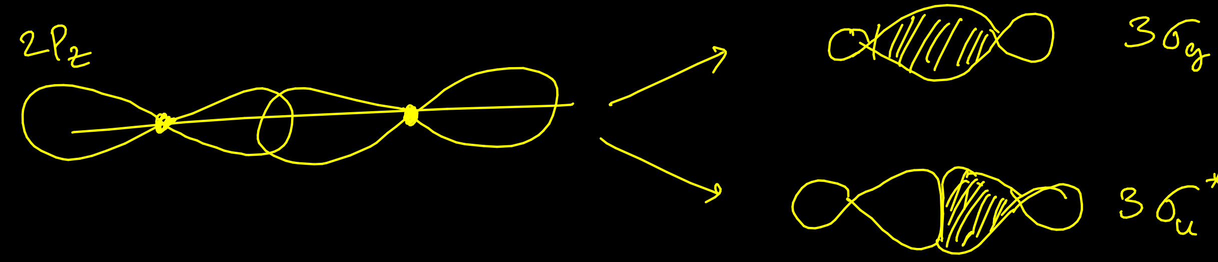
1s



2s

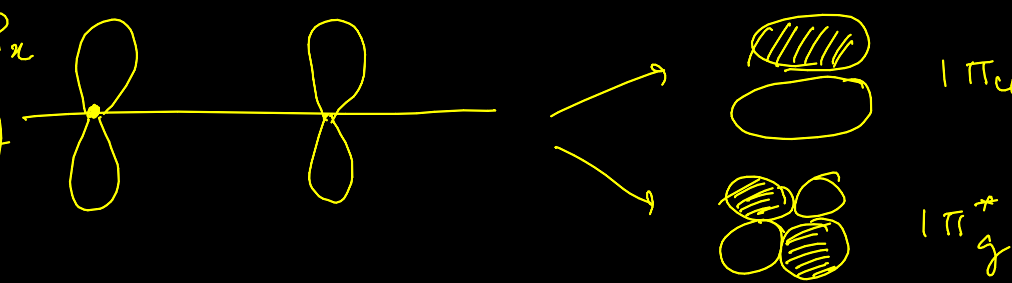


$2P_z$



$2P_x$

$2P_y$



Homonuclear Diatomic + Polyatomic } MO Diagram

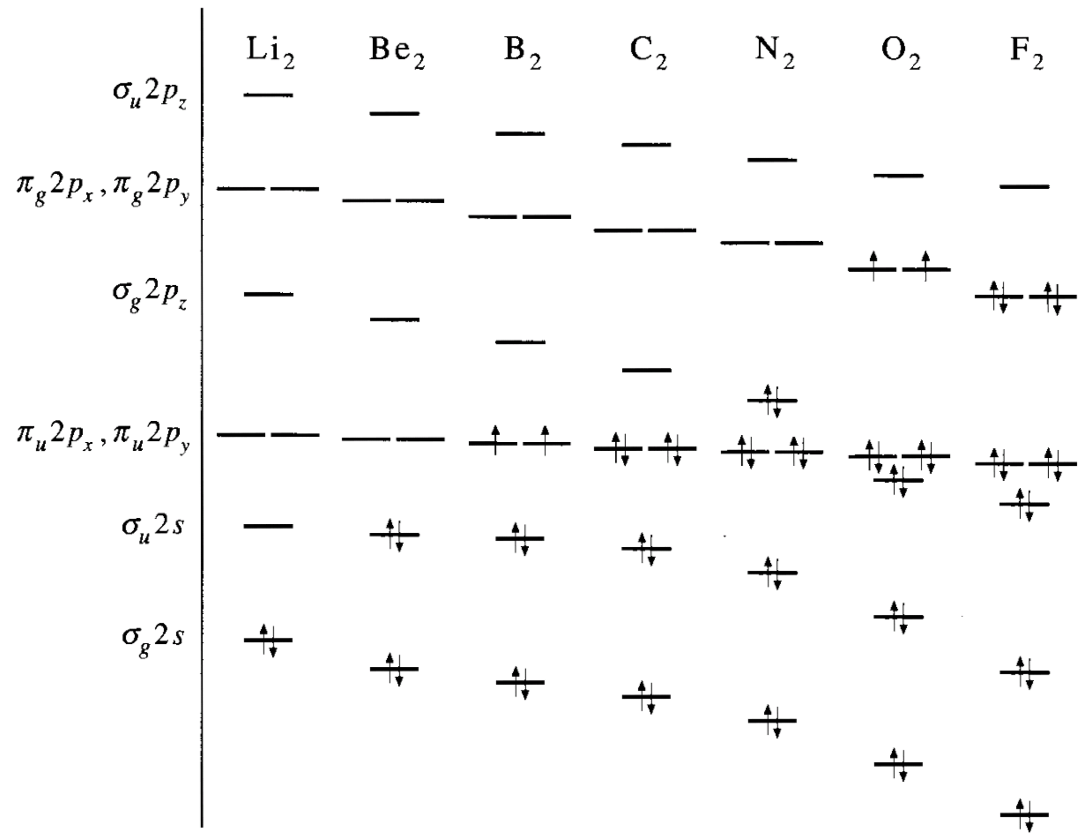


FIGURE 9.13
The relative energies (not to scale) of the molecular orbitals for the homonuclear diatomic molecules Li_2 through F_2. The $\pi_u 2p_x$ and $\pi_u 2p_y$ orbitals are degenerate, as are the $\pi_g 2p_x$ and $\pi_g 2p_y$ orbitals.

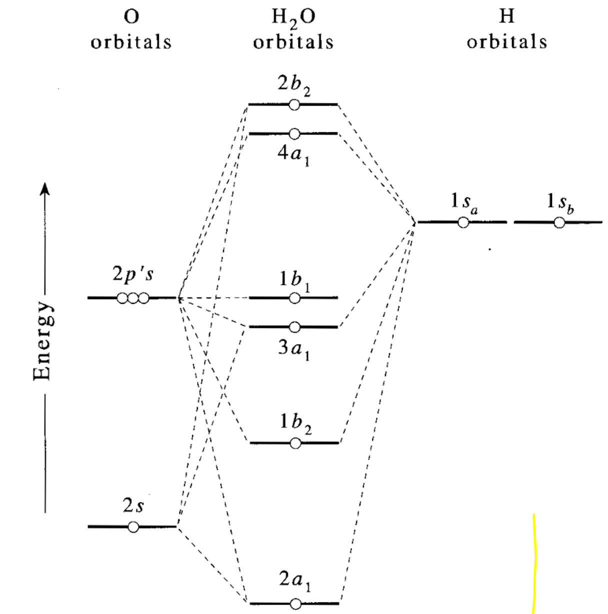


FIGURE 10.13
A molecular-orbital energy-level diagram for the valence electrons in H_2O. (The bond angle is 104.5°, the equilibrium value.) Note that the $1b_1$ orbital is a nonbonding orbital.

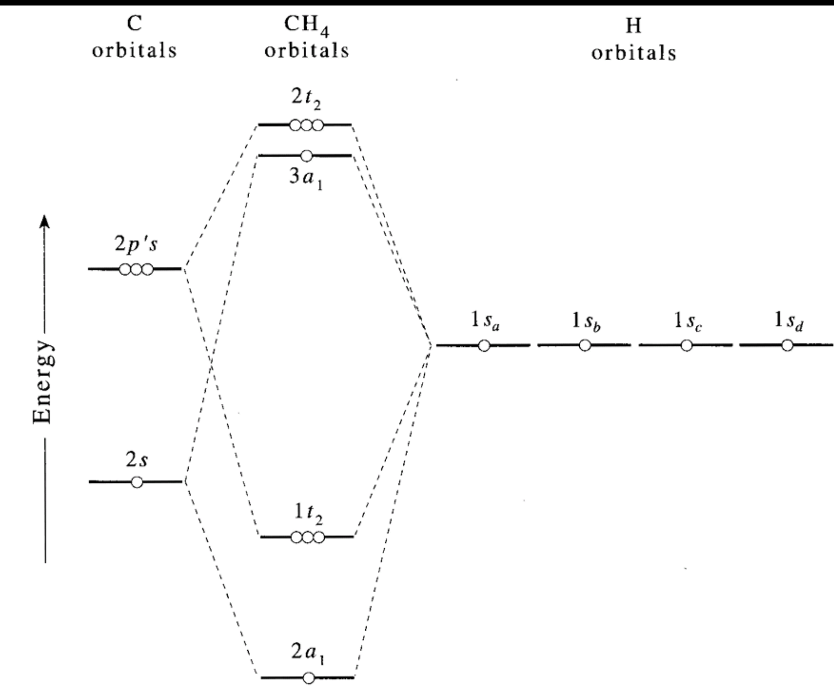
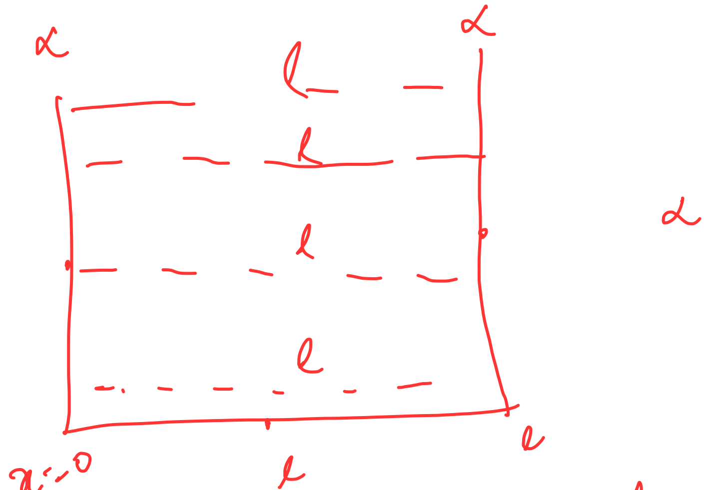
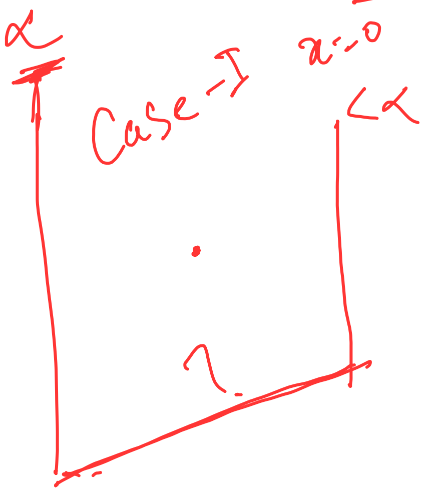
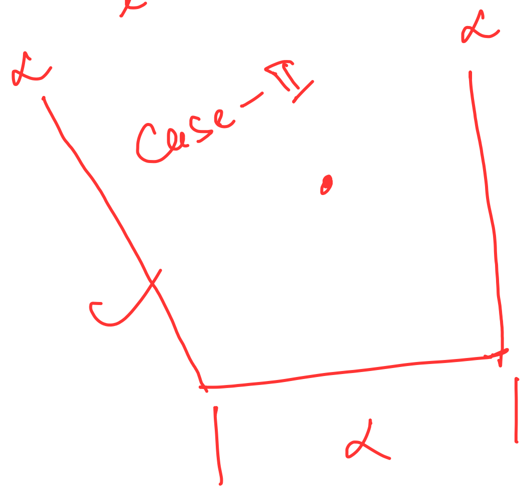


FIGURE 10.15
A molecular-orbital energy-level diagram for the valence electrons in CH_4.







$\varphi^{p}(x)$

$E^{p}(x)$

$$\sqrt{n} = (n-1)!$$